Full Publication List

Preprints and submitted/revised articles

[P1] P. Formont, M. Darrin, I. B. Ayed, and P. Piantanida, "MolRGen: A Training and Evaluation Setting for De Novo Molecular Generation with Reasonning Models," 2026. [paper] [arXiv]

[P2] M. Dubois, F. Yvon, and P. Piantanida, "How much were you told? measuring external information in peer reviews," 2026. [paper] [arXiv]

[P3] P. Piantanida and G. Pichler, "A theory for rational speech acts as KL projections with powered semantics," 2026, (submitted)

[P4] Y. Bahram, Z. Dehghani, M. Desbos, E. Granger, P. Piantanida, and M. Shateri, "Continuous adversarial meanflow transfer," 2026. [paper] [arXiv]

[P5] C. Chataigner, P. Piantanida, and G. Farnadi, "Position: Auditing Is Not Evaluating; LLM Audit Requires Dynamic, Contextual, Budget-Aware and Reliable Evidence," Feb. 2026, working paper or preprint. [paper]

[P6] L. Estienne, E. Ernst, M. Vera, P. Piantanida, and L. Ferrer, "Ecuas$_n$: A family of metrics for principled evaluation of uncertainty-augmented systems," 2026. [paper] [arXiv]

Journal papers

[J1] E. Aubinais, E. Gassiat, and P. Piantanida, "Fundamental Limits of Membership Inference Attacks on Machine Learning Models," *Journal of Machine Learning Research*, vol. 26, no. 263, pp. 1–54, 2025. [paper] [arXiv]

[J2] Z. Dehghani Tafti, P. Piantanida, and M. Shateri, "A Universal Source-Free Class Unlearning Framework via Synthetic Embeddings," *Transactions on Machine Learning Research*, May 2026. [paper]

[J3] P. Morales-Álvarez, S. Christodoulidis, M. Vakalopoulou, P. Piantanida, and J. Dolz, "BayesAdapter: Enhanced Uncertainty Estimation in CLIP Few-Shot Adaptation," *International Journal of Computer Vision*, vol. 134, no. 2, p. 51, February 2026. [paper] [arXiv]

[J4] L. Estienne, M. Vera, E. Fons, E. Kochkina, P. Piantanida, and L. Ferrer, "Adapting Language Models to Produce Good Class Probabilities for Classification Tasks," *Transactions on Machine Learning Research*, February 2026. [paper]

[J5] D. Gorla, L. Jalouzot, F. Granese, C. Palamidessi, and P. Piantanida, "On Estimating the Strength of Differentially Private Mechanisms in a Black-Box Setting," *IEEE Transactions on Dependable and Secure Computing*, vol. 22, no. 5, pp. 5494–5507, September 2025. [paper]

[J6] P. Formont, H. Jeannin, P. Piantanida, and I. Ben Ayed, "A Strong Baseline for Molecular Few-Shot Learning," *Transactions on Machine Learning Research*, February 2025. [paper] [arXiv]

[J7] C. Presvôts, M. Kieffer, T. Prevost, P. Panciatici, Z. Li, and P. Piantanida, "Multiple-model coding scheme for electrical signal compression," *Signal Processing*, vol. 236, p. 110037, November 2025. [paper]

[J8] E. D. C. Gomes, F. Alberge, P. Duhamel, and P. Piantanida, "Combine and Conquer: A Meta-Analysis on Data Shift and Out-of-Distribution Detection," *Transactions on Machine Learning Research*, July 2024. [paper] [arXiv]

[J9] M. Shateri, F. Messina, F. Labeau, and P. Piantanida, "Preserving Privacy in GANs Against Membership Inference Attack," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1728–1743, 2024. [paper] [arXiv]

[J10] G. Pichler, M. Romanelli, L. Rey Vega, and P. Piantanida, "Perfectly Accurate Membership Inference by a Dishonest Central Server in Federated Learning," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 4290–4296, July 2024. [paper] [arXiv]

[J11] M. Vera, L. Rey Vega, and P. Piantanida, "PACMAN: PAC-style bounds accounting for the Mismatch between Accuracy and Negative log-loss," *Information and Inference: A Journal of the IMA*, vol. 13, no. 1, p. iaae002, January 2024. [paper] [arXiv]

[J12] G. Del Grosso, G. Pichler, C. Palamidessi, and P. Piantanida, "Bounding information leakage in machine learning," *Neurocomputing*, vol. 534, pp. 1–17, May 2023. [paper] [arXiv]

[J13] C. Pinzón, C. Palamidessi, P. Piantanida, and F. Valencia, "On the incompatibility of accuracy and equal opportunity," *Machine Learning*, vol. 113, no. 5, pp. 2405–2434, May 2024. [paper]

[J14] M. Vera, L. Rey Vega, and P. Piantanida, "The role of mutual information in variational classifiers," *Machine Learning*, vol. 112, no. 9, pp. 3105–3150, September 2023. [paper] [arXiv]

[J15] M. Picot, F. Granese, G. Staerman, M. Romanelli, F. Messina, P. Piantanida, and P. Colombo, "A Halfspace-Mass Depth-Based Method for Adversarial Attack Detection," *Transactions on Machine Learning Research*, March 2023. [paper]

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[J19] J. F. Silva and P. Piantanida, "On Universal D-Semifaithful Coding for Memoryless Sources With Infinite Alphabets," *IEEE Transactions on Information Theory*, vol. 68, no. 4, pp. 2782–2800, April 2022. [paper] [arXiv]

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[C1] C. Roverato, J. Cotnareanu, P. Piantanida, B. N. Oreshkin, and M. Coates, "DynaPFN: Zero-shot dynamical system forecasting with tabular prior-fitted networks," in *Advances in Neural Information Processing Systems*, 2026

[C2] C. Chataigner, P. Piantanida, and G. Farnadi, "Auditing is not evaluating: LLM audit requires dynamic, contextual, budget-aware and reliable evidence," in *AI4GOOD Workshop at the 40th Conference on Neural Information Processing Systems (NeurIPS 2026)*, 2026. [paper]

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[C4] K. Chehbouni, C. Chataigner, P. Ganesh, P. Piantanida, J. C. K. Cheung, and G. Farnadi, "From infrastructure to interface, the AI value chain drives LLM homogenization," in *Advances in Neural Information Processing Systems*, 2026, neurIPS 2026 Position Paper Track

[C5] A. Satouf, G. D'Erasmo, Y. Zong, H. Amadou-Boubacar, P. Piantanida, and B. Piwowarski, "STORM: Stepwise token optimization with reward-guided beam search," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026. [paper] [arXiv]

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